



Modelling Financial Markets

Chapter 8: Backtesting and Evaluating Risk Forecasts



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ACI: Adaptive Conformal Inference
- AI: Artificial Intelligence
- AR: AutoRegressive (model)
- BCBS: Basel Committee on Banking Supervision
- BET: Bucharest Exchange Trading (index)
- BH: Benjamini–Hochberg (procedure)
- BIS: Bank for International Settlements
- BNR: Banca Națională a României (the National Bank of Romania)
- COVID-19: COronaVirus Disease 2019
- DGP: Data-Generating Process
- DM: Diebold–Mariano (test)
- DQ: Dynamic Quantile (test)
- EGARCH: Exponential GARCH
- ES: Expected Shortfall
- ESR: Expected Shortfall Regression (backtest)
- EVT: Extreme Value Theory
- FDR: False Discovery Rate
- FHS: Filtered Historical Simulation
- FRTB: Fundamental Review of the Trading Book
- FWER: Family-Wise Error Rate
- FZ: Fissler–Ziegel (family of scoring functions for VaR and ES)
- FZ0: Fissler–Ziegel loss, zero-homogeneous version
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- GAS: Generalised Autoregressive Score (model)
- GJR-GARCH: Glosten–Jagannathan–Runkle GARCH

Acronyms used in this material (2/2)

- ▣ GPD: Generalised Pareto Distribution
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HS: Historical Simulation
- ▣ IMA: Internal Models Approach
- ▣ LLM: Large Language Model
- ▣ MCS: Model Confidence Set
- ▣ PIT: Probability Integral Transform
- ▣ POF: Proportion Of Failures (test)
- ▣ POT: Peaks Over Threshold

Outline

Why Backtest Risk Forecasts?

Backtesting VaR: Coverage and Independence

The Regulator's View: Basel Traffic Light and FRTB

Backtesting Expected Shortfall

Elicitability and Scoring Functions

Comparative Backtesting

Conformal Prediction for Tail Risk

Case Study: Backtesting ES with ES Forecasts Only

AI for Scientific Discovery

Which Model Would You Certify?

References

Setup and Notation

- ▣ **Backtesting**: compare risk forecasts made with the information $\mathcal{F}_{t-1}$ with the outcomes that followed
- ▣ Daily log return $X_t = r_t = \ln(P_t/P_{t-1})$; daily **loss** $L_t = -r_t$ (positive number = money lost)
- ▣ **VaR** (Value-at-Risk) and **ES** at tail probability α (Chapter 7)
 - ▶ $\text{VaR}_\alpha(X) = -\inf\{x : P(X \leq x) > \alpha\} = -q_\alpha(X)$; $\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du = E[L_t \mid L_t > \text{VaR}_\alpha]$
(continuous case)
- ▣ Levels are tail probabilities: VaR 1% ($\alpha = 0.01$), VaR 2.5% and ES 2.5% ($\alpha = 0.025$); the target breach rate is α ; Basel used VaR 1% over 10 days, FRTB uses ES 2.5%, backtests use one-day forecasts
- ▣ Plan: coverage and independence tests and their size and power; regulation; ES tests; scoring functions and comparative backtests; conformal prediction

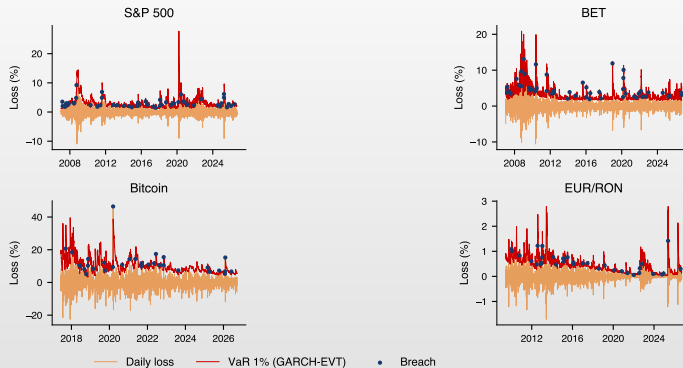
Six Competing Forecasters

- ▣ **HS**: the empirical quantile of order $1 - \alpha$ of the last 1000 losses
- ▣ **Normal**: the Normal distribution with the window mean and standard deviation
- ▣ **Student-t**: a Student-t distribution fitted to the window (location, scale, degrees of freedom ν)
- ▣ **GARCH-t**: $\sigma_t^2 = \omega + a\varepsilon_{t-1}^2 + b\sigma_{t-1}^2$ with standardised Student-t shocks
- ▣ **FHS**: GARCH volatility $\times$ empirical quantiles of the standardised residuals $z_t = \varepsilon_t/\sigma_t$
- ▣ **GARCH-EVT** (McNeil & Frey (2000)): GARCH volatility $\times$ a GPD tail fitted by POT to the largest 10% of standardised losses

The Forecasting Experiment

- ▣ Four series, each on its own trading calendar
 - ▶ S&P 500 and BET (Bucharest Stock Exchange index): closing values
 - ▶ Bitcoin (trades every day) and EUR/RON (BNR reference rate)
- ▣ Rolling window of 1000 observations; each forecast uses only data up to the previous day
- ▣ GARCH, Student-t and EVT parameters re-estimated every 20 days; volatility updated daily
- ▣ Out-of-sample days T (evaluation until 18.09.2026)
 - ▶ S&P 500 4 959 and BET 4 937 (from 2007), EUR/RON 4 352 (from 2009), Bitcoin 3 384 (from 2017)

Four Markets, One Risk Model



Orange: daily losses; red: VaR 1% forecast by GARCH-EVT; blue dots: breaches ($L_t > VaR_t$)

Four Markets: What We See

- ▣ The VaR moves with volatility: it doubles or triples in 2008, March 2020 and the 2022 crypto crash
- ▣ Breach rates of the GARCH-EVT VaR 1%
 - ▶ S&P 500 1.13%, BET 0.99%, Bitcoin 0.95%, EUR/RON 0.78%
- ▣ EUR/RON has a very different scale: a managed exchange rate with long calm periods and rare jumps
- ▣ Question for the chapter: is a rate close to 1% enough to call a model good?

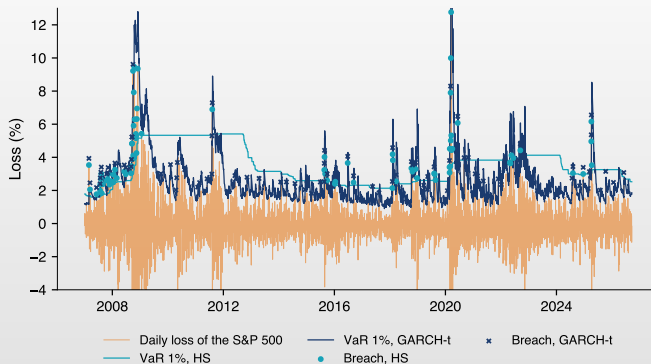
What a Backtest Can and Cannot Identify

- Ideal conditional calibration: $P(L_t > \text{VaR}_t \mid \mathcal{F}_{t-1}) = \alpha$ for all t
 - ▶ Each backtest checks one implication: Kupiec the unconditional rate, Christoffersen first-order memory, DQ predictability from chosen regressors
 - ▶ One outcome per day, so evidence accumulates only over many days
- H_0 is a **joint** hypothesis on the whole forecasting method
 - ▶ distribution, volatility dynamics, estimated parameters, window length and refit rule
 - ▶ A rejection does not say which component failed
- The textbook χ^2 limits treat VaR_t as known; with estimated parameters the hits are not exactly Bernoulli(α) even under a correct model (estimation risk, end of Section 2)
- Non-rejection is not validation: it is informative only if the test has power against the relevant alternatives at the available T

The Hit Sequence

- A **breach** (exception, violation) occurs when the loss exceeds the forecast: $I_t = 1\{L_t > \text{VaR}_t^\alpha\}$
- A correct VaR implies two properties (Christoffersen (1998))
 - ▶ **Unconditional coverage:** $P(I_t = 1) = \alpha$, so about αT breaches in T days
 - ▶ **Independence:** I_t is independent of the whole past hit history (Christoffersen's test checks only first-order Markov memory)
- Together: I_t are i.i.d. Bernoulli(α) variables
- A model can fail either way: too many breaches, or breaches that come in clusters

S&P 500: Two VaR Forecasts, 2007–2026

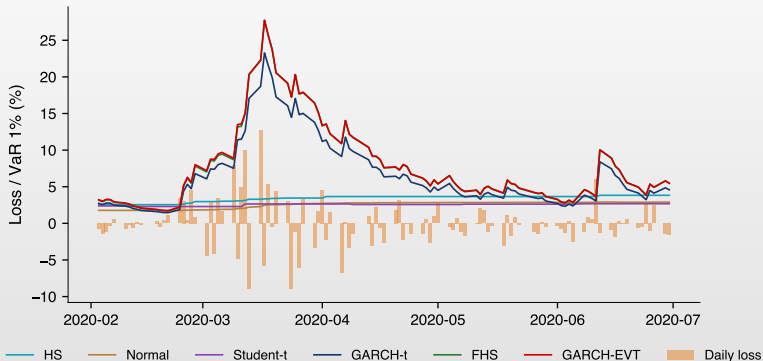


□ VaR 1%: HS in teal, GARCH-t in blue; orange: daily losses; markers: breaches of each model

HS versus GARCH-t: Interpretation

- HS reacts slowly: its VaR is a step function that jumps only when extreme days enter or leave the window
- GARCH-t reacts the next day: the VaR rises as soon as volatility rises
- Almost the same number of breaches: HS 83, GARCH-t 88 in 4 959 days
 - ▶ but HS breaches arrive in bursts (2008, 2020), GARCH-t breaches are spread over time
- Counting breaches is therefore not enough: we must also test **when** they occur

March 2020: The Stress Test Nobody Planned



□ S&P 500, February–June 2020: daily losses (bars) and the VaR 1% of the six models

March 2020: Interpretation

- Largest loss 12.77% on 16.03.2020; in 104 days the expected number of breaches of a correct VaR 1% is 1.04
- Breaches in these 104 days
 - ▶ Unconditional models: HS 12, Normal 17, Student-t 17
 - ▶ Conditional models: GARCH-t 3, FHS 2, GARCH-EVT 2
- A fat-tailed distribution (Student-t) does not help if the scale does not move with volatility
- Lesson: in a crisis, conditioning on current volatility matters more than the shape of the tail

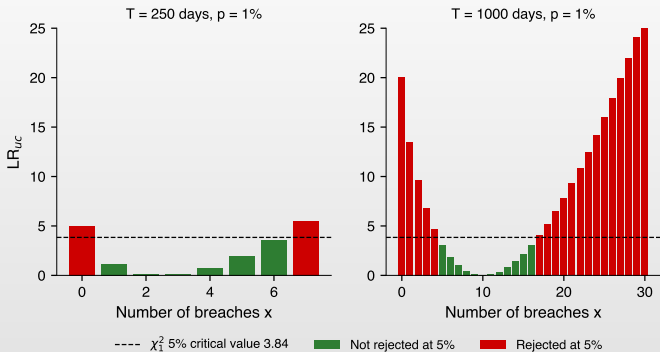
Kupiec (1995): the POF Test

- ▣ Data: $x = \sum_t I_t$ breaches in T days; estimated breach rate $\hat{\pi} = x/T$
- ▣ $H_0: \pi = \alpha$ (correct unconditional coverage), against $H_1: \pi \neq \alpha$
- ▣ Likelihood ratio statistic (Kupiec (1995))
 - ▶ $LR_{uc} = -2 \ln \frac{(1-\alpha)^{T-x} \alpha^x}{(1-\hat{\pi})^{T-x} \hat{\pi}^x} \xrightarrow{d} \chi_1^2$
- ▣ Reject at 5% if $LR_{uc} > 3.84$; the test is two-sided: too few breaches also count as failure
- ▣ Exact alternative for small T : the Binomial(T, α) distribution of x

Exact Test and Local Power of the POF Test

- ▣ S&P 500, GARCH-t, 2008: $x = 8$ in $T = 253$ (expected 2.53): $LR_{uc} = 7.60$, $p = 0.0058$; exact Binomial $p = 2P(X \geq 8) = 0.009$
- ▣ Local alternatives $\pi_T = \alpha + h/\sqrt{T}$: a second-order expansion of the Bernoulli log-likelihood gives
 - ▶ $LR_{uc} = \frac{T(\hat{\pi} - \alpha)^2}{\alpha(1 - \alpha)} + o_p(1) \xrightarrow{d} \chi_1^2(\lambda)$, $\lambda = \frac{h^2}{\alpha(1 - \alpha)} = \frac{T(\pi - \alpha)^2}{\alpha(1 - \alpha)}$
- ▣ $T = 250$, $\pi = 2\%$: $\lambda = 2.53$, local power 36%, exact power 24%: with $\alpha T = 2.5$ the discreteness of x dominates
- ▣ Exact size at $T = 250$ is 9.5%, not 5%: $P(x = 0) = 0.99^{250} = 8.1\%$ already falls in the rejection region

How Many Breaches Are Acceptable?

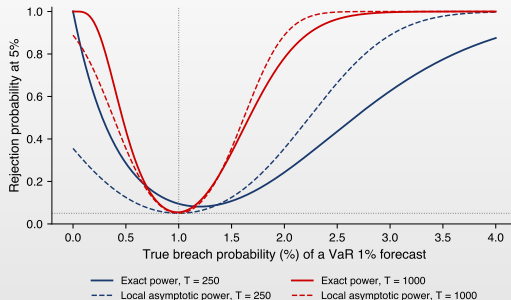


□ LR_{uc} as a function of the number of breaches x , for one year (left) and four years (right) of VaR 1%

The Low Power of Coverage Tests

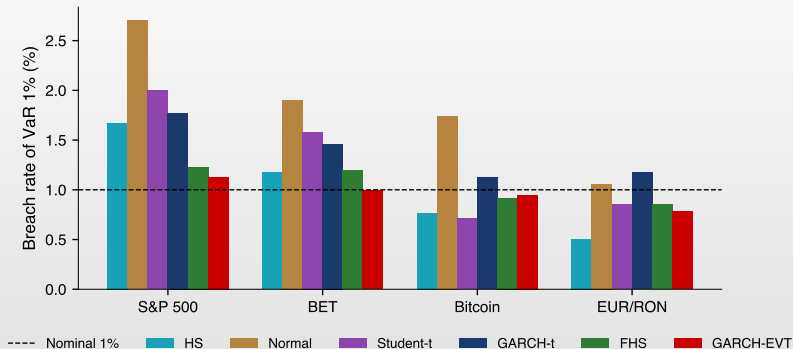
- ▣ With $T = 250$ the test accepts x from 1 to 6: a model with a true breach rate of 2% (double the target) passes with probability 76%
- ▣ Even $x = 0$ is rejected ($LR_{uc} = -500 \ln 0.99 = 5.03$): a VaR that is never breached is too conservative
- ▣ With $T = 1000$ the acceptance region narrows to 5–16 breaches
- ▣ Tail events are rare by construction: one year of data contains about 2.5 informative days for VaR 1%
- ▣ Consequence: long samples, several levels ($\alpha = 1\%$ and 2.5%) and several tests are needed

Power of the POF Test



- Rejection probability of the Kupiec test at 5% as a function of the true breach probability; solid: exact Binomial; dashed: local $\chi^2_1(\lambda)$ approximation
- 50% power needs $\pi \approx 2.7\%$ at $T = 250$ and $\pi \approx 1.7\%$ at $T = 1000$; the gap between the exact and local curves reflects the discrete rejection region

Breach Rates Across Models and Markets



□ Share of days with $L_t > \text{VaR}_t$ for the VaR 1%; dashed line: the nominal 1%

Breach Rates: Interpretation

- ▣ The Normal distribution fails everywhere except EUR/RON: S&P 500 2.70%, BET 1.90%, Bitcoin 1.74%
- ▣ GARCH-t is breached too often on equities (S&P 500 1.77%, BET 1.46%); FHS, with the same GARCH-t volatility, is much closer to 1% (1.23%, 1.20%): evidence against the symmetric t tail on the loss side
- ▣ FHS and GARCH-EVT stay closest to 1% in all four markets
- ▣ HS is too conservative on EUR/RON (0.51%): an old crisis in the window keeps its VaR high
- ▣ Kupiec p -values (S&P 500): HS < 0.001 , GARCH-t < 0.001 , FHS 0.116, GARCH-EVT 0.370

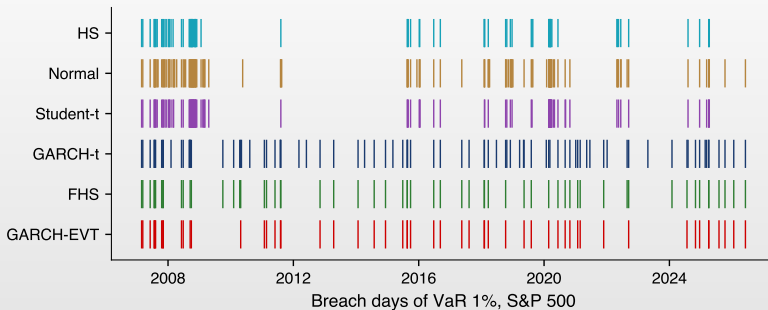
Christoffersen (1998): Testing Independence

- ▣ Model I_t as a first-order Markov chain with $\pi_{ij} = P(I_t = j \mid I_{t-1} = i)$
- ▣ Count transitions n_{ij} (from state i yesterday to state j today); $\hat{\pi}_{01} = \frac{n_{01}}{n_{00} + n_{01}}$, $\hat{\pi}_{11} = \frac{n_{11}}{n_{10} + n_{11}}$
- ▣ H_0 : $\pi_{01} = \pi_{11} = \pi$ (no memory), with $\hat{\pi} = (n_{01} + n_{11}) / \sum n_{ij}$
- ▣ Statistic
 - ▶
$$\text{LR}_{ind} = -2 \ln \frac{(1 - \hat{\pi})^{n_{00} + n_{10}} \hat{\pi}^{n_{01} + n_{11}}}{(1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \xrightarrow{d} \chi_1^2$$
- ▣ Conditional coverage: $\text{LR}_{cc} = \text{LR}_{uc} + \text{LR}_{ind} \xrightarrow{d} \chi_2^2$ tests both properties at once

HS in 2007–2010: Why the Markov Test Misses

- ▣ S&P 500, VaR 1% of HS: $n_{00} = 928$, $n_{01} = 38$, $n_{10} = 38$, $n_{11} = 3$; $\hat{\pi}_{01} = 0.0393$, $\hat{\pi}_{11} = 0.0732$
- ▣ The breach probability almost doubles after a breach, yet $LR_{ind} = 0.95$ ($p = 0.33$); coverage fails: $LR_{cc} = 55.19$
- ▣ $\hat{\pi}_{11}$ rests on 3 consecutive pairs: its standard error is of order $\sqrt{\hat{\pi}_{11}(1 - \hat{\pi}_{11})/(n_{10} + n_{11})}$, larger than the effect
- ▣ Clusters spanning several days escape a first-order chain: we need tests that use more lags or the whole spacing pattern

When Do Breaches Happen?



- Each tick is one breach of the VaR 1% on the S&P 500

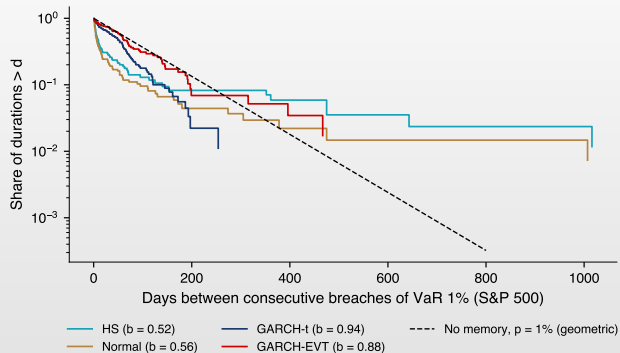
Breach Clustering: Interpretation

- ▣ HS, Normal and Student-t: dense bursts in 2008, 2015–2016, 2018, 2020 and 2022, and long silent periods
- ▣ GARCH-t, FHS and GARCH-EVT: breaches spread almost uniformly over 20 years
- ▣ Clusters mean that losses come **exactly when capital is most needed**
- ▣ Independence p -values (S&P 500): HS <0.001 , Normal <0.001 , GARCH-t 0.095, FHS 0.046

Duration-Based Tests

- ▣ Duration D_i : number of days between breach $i - 1$ and breach i
- ▣ Under a correct VaR, breaches have no memory: D_i is geometric with mean $1/\alpha$ (100 days for $\alpha = 1\%$)
- ▣ Christoffersen & Pelletier (2004): approximate the discrete durations by a continuous Weibull model
 - ▶ $f(D; a, b) = a^b b D^{b-1} e^{-(aD)^b}$; $b = 1$ is the exponential (memoryless) case
 - ▶ $b < 1$: short durations are too frequent, i.e. breaches cluster
- ▣ H_0 : $b = 1$ with a free, a likelihood ratio test (χ^2_1) of memorylessness, not of the mean $1/\alpha$ (coverage is left to Kupiec); first and last durations censored
- ▣ Uses the whole spacing pattern, not only consecutive days: more power against clustering

How Long Between Breaches?



□ Share of durations longer than d (log scale); dashed: no memory with $\alpha = 1\%$

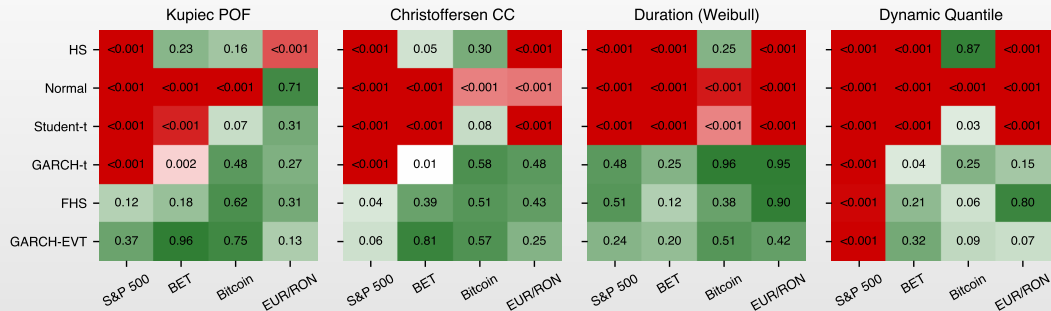
Durations: Interpretation

- ▣ HS: $b = 0.52$, median duration 7 days, 44% of durations are at most 5 days
- ▣ Normal: $b = 0.56$; GARCH-t: $b = 0.94$ (median 48 days); GARCH-EVT: $b = 0.88$
- ▣ Duration test p -values: HS < 0.001 , GARCH-t 0.479, GARCH-EVT 0.240
- ▣ The duration test rejects HS decisively, where the Markov test for 2007–2010 did not
- ▣ Long tails of HS durations: years without breaches after a crisis leaves a high VaR in the window

Regression-Based Backtests: the DQ Test

- Under correct conditional coverage $\text{Hit}_t = I_t - \alpha$ is a martingale difference: $E[\text{Hit}_t | \mathcal{F}_{t-1}] = 0$, $\text{Var}(\text{Hit}_t | \mathcal{F}_{t-1}) = \alpha(1 - \alpha)$
- So no $\mathcal{F}_{t-1}$ -measurable variable may predict it: regress $\text{Hit}_t = X_t\beta + u_t$ with $X_t = (1, \text{Hit}_{t-1}, \dots, \text{Hit}_{t-4}, \text{VaR}_t)$ and test $\beta = 0$
- Dynamic Quantile statistic ([Engle & Manganelli \(2004\)](#)), Wald form with the null variance)
 - ▶ $\text{DQ} = \frac{\hat{\beta}' X' X \hat{\beta}}{\alpha(1 - \alpha)} = \frac{\text{Hit}' X (X' X)^{-1} X' \text{Hit}}{\alpha(1 - \alpha)} \xrightarrow{d} \chi_6^2$
- Nests Kupiec ($X_t = 1$ gives the score form $T(\hat{\pi} - \alpha)^2 / (\alpha(1 - \alpha))$) and Christoffersen (one lag); adds power against memory beyond lag 1 and against hits that depend on the VaR level
- Extensions: binary logit/probit hit regressions

All VaR Tests at a Glance



□ p -values of the Kupiec, conditional coverage, duration and DQ tests, VaR 1%; red: rejection, green: no rejection

VaR Tests: Interpretation

- Unconditional models (HS, Normal, Student-t) fail the duration test in every market, with one exception: HS on Bitcoin ($p = 0.246$)
- GARCH-EVT is never rejected at 5% by the coverage and duration tests on BET, Bitcoin and EUR/RON
- DQ rejects every model on the S&P 500, including FHS ($p < 0.001$) and GARCH-EVT ($p < 0.001$): hits are predictable from the VaR level and lagged hits, a conditional failure that the counts miss
- On BET and EUR/RON the filtered models pass DQ (FHS: $p = 0.208$ and 0.800); Bitcoin is the easiest market, since every 1000-day window already contains extreme moves
- Several tests on the same breaches are not independent: read the pattern, not a single p -value

Backtests Under Estimation Risk

- ▣ Forecasts use $\hat{\theta}$ estimated on R days; the test uses P out-of-sample days. The hits $\hat{I}_t = 1\{L_t > \text{VaR}_t(\hat{\theta})\}$ are not those of the true θ_0
- ▣ Fixed scheme, expansion ([Escanciano & Olmo \(2010\)](#)):

$$\frac{1}{\sqrt{P}} \sum_t (\hat{I}_t - \alpha) = \frac{1}{\sqrt{P}} \sum_t (I_t - \alpha) + \sqrt{P/R} D' \sqrt{R} (\hat{\theta} - \theta_0) + o_p(1)$$
 - $D = E[\partial P(L_t > \text{VaR}_t(\theta) | \mathcal{F}_{t-1}) / \partial \theta]$ at θ_0 : the second term vanishes only if $P/R \rightarrow 0$
- ▣ This chapter uses a rolling scheme ($R = 1000$, refit every 20 days, $P \approx 5000$): the correction term depends on the scheme, but with $P/R \approx 5$ the χ^2 critical values are not justified either
- ▣ Illustration with the fixed scheme (not our rolling rule): GARCH(1,1)-t with the S&P 500 estimates, $R = 1000$, 400 paths: the Kupiec test rejects a **correct** model in 27.5% of samples at $P = 5000$ (4.0% with the true parameters)
- ▣ Remedies: the corrected variance or subsampling of [Escanciano & Olmo \(2010\)](#), or a parametric bootstrap that re-estimates the model on every simulated path

Size and Power of VaR Backtests: a Monte Carlo

- DGP: GARCH(1,1)-t fitted to the S&P 500 1990–2026 ($a = 0.102$, $b = 0.892$, $\nu = 6.1$); VaR 1%; 1000 paths; rejection rates (%) at 5%

Forecaster	T	POF	CC	Duration	DQ
Correct (size)	250	9.8	1.2	14.8	8.6
	1000	6.5	3.3	7.9	11.0
	5000	4.9	3.4	4.6	8.8
Normal-GARCH (power)	250	10.5	4.9	14.7	18.8
	1000	30.3	19.9	8.0	30.3
	5000	83.8	77.2	7.2	70.7
HS, 1000 days (power)	250	51.2	17.2	12.3	32.4
	1000	46.3	47.1	68.5	73.8
	5000	57.1	86.4	100.0	99.5

- Correct: true parameters and σ_t ; Normal-GARCH: true σ_t , Normal tail; duration p undefined (fewer than 3 durations) counted as non-rejection; Monte Carlo standard error 0.7 points at 5%, 1.6 at 50%

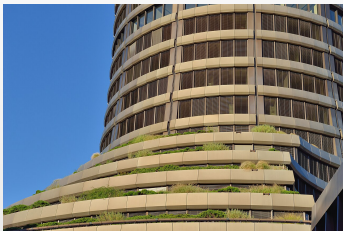
Size and Power: Interpretation

- Size at $T = 250$ is far from 5%: POF 9.8%, CC 1.2%, duration 14.8% (undefined in 28% of samples), DQ 8.6%
- A Normal tail on the right volatility is almost undetectable in one year (POF 10.5%) and needs $T = 5000$ for 83.8% power; durations never see it (7.2%): the misspecification has no memory
- HS on GARCH data: durations and DQ reach 100.0% and 99.5% at $T = 5000$, but only 12.3% and 32.4% in one year
- DQ stays oversized even at $T = 5000$ (8.8%): with rare hits the χ^2_6 approximation is poor; simulate critical values
- Lesson: report the size and power of the test battery at the sample size used, not only the p -values

Recap: Backtesting VaR

- ▣ Kupiec: right number of breaches; Christoffersen: no first-order memory; Christoffersen–Pelletier: no memory in durations
- ▣ DQ regression: predictability of hits from lags and the VaR level; estimated parameters with P/R large distort every χ^2 limit
- ▣ Coverage tests have low power and wrong size in one year; durations detect the clustering that coverage misses
- ▣ Empirically: conditional models (GARCH filter) beat unconditional ones; the tail model (FHS, EVT) fixes the remaining excess

The Basel Committee and Backtesting



BIS tower, Basel: seat of the BCBS secretariat

Photo: Taxiarchos228 (2014); Free Art License; Wikimedia

[Commons](#)

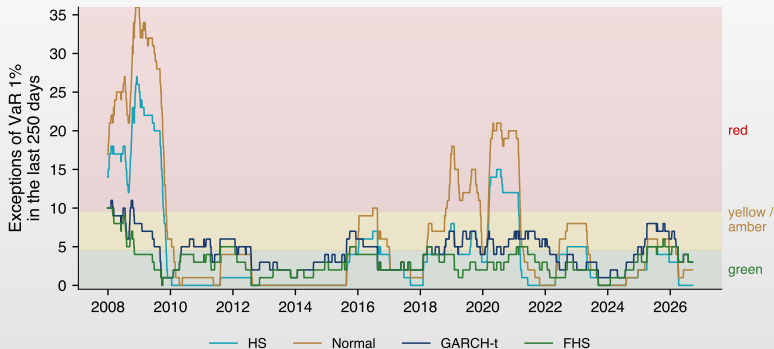
- Banks using internal models must backtest their daily VaR 1% against actual P&L
- BCBS (1996) introduced the **traffic light**: exceptions in the last 250 days determine a capital add-on
- BCBS (2019) (FRTB) keeps the idea, but capital is now based on ES 2.5%
- The regulator wants a rule that is simple, automatic and hard to game

The Traffic Light: Where Do the Zones Come From?

Exceptions	$P(X \leq x)$	Zone
0	8.11%	green
2	54.32%	green
4	89.22%	green
5	95.88%	yellow
7	99.60%	yellow
9	99.97%	yellow
10	99.99%	red

- ▣ $X \sim \text{Binomial}(250, 0.01)$: exceptions of a correct VaR 1% in 250 days
- ▣ Yellow starts where $P(X \leq x) \geq 95\%$, red where $P(X \leq x) \geq 99.99\%$ (BCBS (1996))
- ▣ 1996 rule: the capital multiplier 3 rises by 0.40–0.85 in the yellow zone and by 1 in the red zone
- ▣ Only the **number** of exceptions counts, not their size or timing

The Traffic Light Through Time



□ Exceptions of the VaR 1% in the last 250 days, S&P 500; shaded: green, yellow and red zones

Traffic Light: Interpretation

- ▣ Share of days in the red zone: Normal 22.3%, HS 14.7%, GARCH-t 2.0%, FHS 0.8%
- ▣ Peak: HS 27 and Normal 37 exceptions in the 250 days ending 01.12.2008
- ▣ GARCH-t spends 46.6% of the time in the yellow zone: too many exceptions, but never many at once
- ▣ The rule punishes models **after** a crisis, when capital is already strained: backtesting is backward-looking

FRTB: What Changed in 2019

- ▣ Capital is computed from ES 2.5%, which sees the size of tail losses (BCBS (2019))
- ▣ Backtesting is still done on VaR, at two levels
 - ▶ Bank-wide: VaR 1%, 250 days; multiplier 1.50 (green, 0–4), 1.70–1.92 (amber, 5–9), 2.00 (red, 10 or more)
 - ▶ Trading desk: more than 12 exceptions of VaR 1% or 30 of VaR 2.5% in 12 months sends the desk to the standardised approach
- ▣ Why backtest VaR if capital uses ES? ES alone is not elicitable (Section 5), and VaR tests are simple and robust
- ▣ IMA approval also requires a profit-and-loss attribution test (not covered here)

FRTB Desk Check on the Last 250 Days

Model	S&P 500	BET	Bitcoin	EUR/RON
HS	0 / 2	4 / 7	3 / 4	1 / 5
Normal	2 / 2	6 / 9	3 / 5	1 / 4
Student-t	0 / 2	4 / 9	1 / 4	5 / 8
GARCH-t	3 / 6	4 / 9	3 / 5	4 / 5
FHS	3 / 4	4 / 6	4 / 5	1 / 4
GARCH-EVT	3 / 3	2 / 7	4 / 5	1 / 4

- Cells: exceptions of VaR 1% / of VaR 2.5% in the last 250 days of each series (ending 18.09.2026)
- Limits: 12 for VaR 1%, 30 for VaR 2.5%; every model passes comfortably
- Illustration only: FRTB applies the limits to desk P&L (actual and hypothetical), not to index returns
- The last year was calm: a green light today says little about the next crisis

Recap: The Regulator's View

- Traffic light = a Binomial test with fixed thresholds, calibrated for 250 days of VaR 1%
- FRTB: capital on ES 2.5%; backtesting on VaR 1% bank-wide and on both VaR 1% and VaR 2.5% per trading desk
- Unconditional models would have spent long periods in the red zone after 2008 and 2020

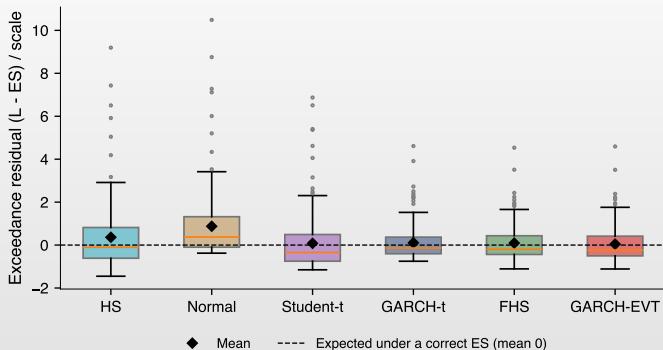
Why ES Is Harder to Backtest

- ▣ A VaR forecast is checked with a yes/no event per day; ES is an **average** over the tail beyond VaR
- ▣ Only the breach days carry information: about αT days (124 for the S&P 500 with $\alpha = 2.5\%$)
- ▣ In 250 days there are $250 \cdot 0.025 = 6.25$ tail observations; the precision of the tail mean scales like $1/\sqrt{\alpha T} \approx 0.40$
- ▣ [Pele & Mazurencu-Marinescu-Pele \(2026\)](#): roughly one in five pairwise ES comparisons in global markets is precision-fragile
- ▣ Three families of tests: residual tests, Acerbi–Szekely statistics, cumulative violations

McNeil and Frey (2000): Exceedance Residuals

- On each breach day of the VaR 2.5% compute $e_t = (L_t - \text{ES}_t)/\sigma_t$, with σ_t the forecast volatility
- If VaR **and** ES are correct given $\mathcal{F}_{t-1}$, and $\sigma_t > 0$ is known at $t - 1$: $E[e_t \mid L_t > \text{VaR}_t, \mathcal{F}_{t-1}] = 0$; a wrong VaR changes the conditioning event
- H_0 : $E[e_t] = 0$ against H_1 : $E[e_t] > 0$ (ES underestimated); one-sided t -statistic
- Few and fat-tailed residuals: [McNeil & Frey \(2000\)](#) obtain the p -value with an i.i.d. bootstrap of the centred residuals, which assumes independent residuals (use a block bootstrap otherwise)
- Scaling by σ_t makes residuals comparable between calm and turbulent days

Exceedance Residuals on the S&P 500



□ Distribution of $(L_t - ES_t) / \sigma_t$ on breach days of the VaR 2.5%; diamonds: means

Exceedance Residuals: Interpretation

- ▣ Means: Normal 0.875, HS 0.363, GARCH-t 0.106, FHS 0.093, GARCH-EVT 0.045
- ▣ Bootstrap p -values: Normal <0.001 , HS <0.001 , GARCH-t 0.023, FHS 0.096, GARCH-EVT 0.246
- ▣ The Normal distribution underestimates the tail mean by almost one volatility unit
- ▣ The outliers (up to 10 volatility units) are the days that decide capital adequacy

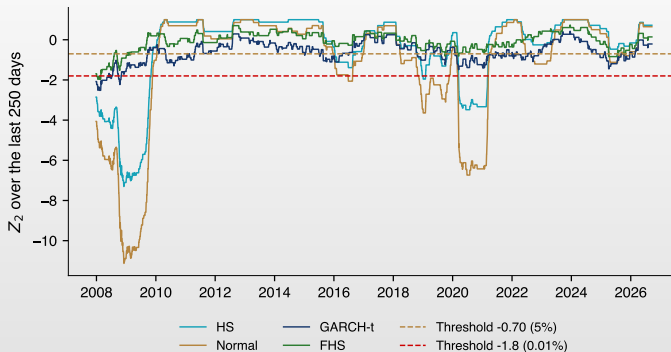
Acerbi and Szekely (2014): Testing ES Directly

- ▣ Breach indicator at the ES level: $I_t = 1\{L_t > \text{VaR}_t^\alpha\}$, $N_T = \sum_t I_t$, $\alpha = 0.025$
- ▣ $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t I_t}{\text{ES}_t}$: is the average tail loss as large as forecast? (conditional on breaches)
- ▣ $Z_2 = 1 - \frac{1}{T_\alpha} \sum_t \frac{L_t I_t}{\text{ES}_t}$: tests frequency and size together
- ▣ Under H_0 (correct forecast distribution) $E[Z_2] = 0$; underestimated risk gives $Z_2 < 0$ (Acerbi & Szekely (2014))
- ▣ For $T = 250$ and many distributions, the 5% and 0.01% critical values are close to -0.70 and -1.8 : a traffic light for ES

Getting p -Values by Simulation

- The null distribution of Z_2 depends on the forecast distributions F_t , which the model supplies
- Algorithm (repeat $M = 2000$ times)
 - ▶ For each day draw $U_t \sim \text{Uniform}(0,1)$ and set $L_t^* = F_t^{-1}(U_t)$
 - ▶ Compute Z_2^* from the simulated losses with the same VaR and ES forecasts
- p -value = share of simulated Z_2^* at or below the observed Z_2
- Only tail draws matter: for HS and FHS we store the 25 largest window losses of each day
- S&P 500, full sample: $Z_2 = -0.618$ for GARCH-t ($p < 0.001$) versus -0.091 for FHS ($p = 0.172$); critical value about -0.152

Z_2 Over Rolling 250-Day Windows



□ Z_2 of the ES 2.5% forecasts over the last 250 days, S&P 500, with the fixed thresholds -0.70 and -1.8

Z_2 Traffic Light: Interpretation

- Minimum Z_2 : Normal -11.13 and HS -7.30 (window ending 01.12.2008); GARCH-t -2.52 ; FHS -1.97
- Share of windows below -1.8 (red): Normal 21.9%, HS 15.8%, GARCH-t 3.0%, FHS 0.7%
- GARCH-t sits below -0.70 in 39.6% of windows: frequent small underestimation of the tail
- Z_2 punishes the size of the misses, not only their number: HS falls to -7.30 while its exception count peaked at 27

Du and Escanciano (2017): Cumulative Violations

- ▣ PIT: $u_t = F_t(L_t)$, uniform on $(0, 1)$ if the forecast distribution is correct
- ▣ Cumulative violation: $H_t = \frac{1}{\alpha} (u_t - (1 - \alpha)) 1\{u_t > 1 - \alpha\}$: how deep into the tail the loss went
- ▣ Under H_0 : $E[H_t] = \alpha/2$ and $\text{Var}(H_t) = \alpha(1/3 - \alpha/4)$
- ▣ Unconditional test: $U = \sqrt{T} (\bar{H} - \alpha/2) / \sqrt{\alpha(1/3 - \alpha/4)} \xrightarrow{d} N(0, 1)$
- ▣ Conditional test: Box–Pierce statistic on the first 5 autocorrelations of H_t , χ^2_5 (Du & Escanciano (2017))

All ES Tests at a Glance

	McNeil-Frey				Acerbi-Szekely Z2				Du-Escanciano U			
HS	<0.001	0.48	0.70	0.46	<0.001	0.13	0.86	1.00	<0.001	0.36	0.14	<0.001
Normal	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.001	0.97	<0.001	<0.001	<0.001	0.25
Student-t	0.17	0.01	1.00	0.47	0.002	0.003	0.62	0.92	<0.001	<0.001	0.93	0.05
GARCH-t	0.02	0.04	0.50	0.41	<0.001	<0.001	0.05	0.63	<0.001	<0.001	0.13	0.86
FHS	0.10	0.63	0.60	0.75	0.17	0.19	0.35	0.97	0.09	0.46	0.95	0.09
GARCH-EVT	0.25	0.57	0.29	0.69	0.09	0.21	0.58	0.98	0.24	0.54	0.70	0.11
	S&P 500	BET	Bitcoin	EUR/RON	S&P 500	BET	Bitcoin	EUR/RON	S&P 500	BET	Bitcoin	EUR/RON

□ p -values for ES 2.5%: McNeil-Frey (bootstrap), Z_2 (simulation), Du-Escanciano U

ES Tests: Interpretation

- ▣ The Normal distribution is rejected by all three tests on S&P 500, BET and Bitcoin
- ▣ FHS and GARCH-EVT pass the three ES tests above at 5% on all four markets (the multinomial test below is stricter)
- ▣ GARCH-t fails Z_2 on equities ($p < 0.001$ on S&P 500, < 0.001 on BET); FHS, with the same volatility filter, passes: the failure points to the t tail
- ▣ EUR/RON: most models over-predict ES ($Z_2 > 0$), a consequence of the managed exchange rate
- ▣ The tests agree on the ranking but not on marginal cases: Student-t on the S&P 500 has McNeil–Frey $p = 0.173$ but Z_2 $p = 0.002$

Multinomial VaR Backtests: an Implicit ES Test

- ES 2.5% averages the quantiles beyond VaR 2.5%: test several VaR levels at once (Kratz, Lok & McNeil (2018))
- $N = 4$ tail levels $\alpha_j = 2.5\% (1 - (j - 1)/4)$: 2.5%, 1.875%, 1.25%, 0.625%; each day falls in one of 5 cells via the PIT u_t
- Pearson statistic $S = \sum_{j=0}^4 (O_j - T p_j)^2 / (T p_j) \xrightarrow{d} \chi_4^2$, cell probabilities $p_0 = 97.5\%$, $p_1 = \dots = p_4 = 0.625\%$

p -value	S&P 500	BET	Bitcoin	EUR/RON
HS	<0.001	0.725	0.564	0.004
Normal	<0.001	<0.001	<0.001	<0.001
Student-t	<0.001	0.024	0.015	0.172
GARCH-t	<0.001	<0.001	0.227	0.568
FHS	0.008	0.846	0.413	0.392
GARCH-EVT	0.287	0.864	0.969	0.240

- Only GARCH-EVT passes in all four markets; S&P 500 GARCH-t cells (4767, 36, 46, 50, 60) against 31 expected in each tail cell: too many days in every tail cell, most in the deepest

Two Recent ES Backtests: Regression and Monitoring

- ▣ ES regression backtest (Bayer & Dimitriadis (2022))
 - ▶ Regress L_t on $(1, ES_t)$ in a joint (quantile, ES) regression; H_0 : intercept 0, slope 1
 - ▶ Needs only the ES forecasts, not the forecast distribution: its advantage over Z_2 , whose p -values need simulation
- ▣ Sequential monitoring (Hoga & Demetrescu (2023))
 - ▶ A regulator checks the forecasts every day: repeated fixed-sample tests inflate the overall size
 - ▶ Monitoring statistics with boundary functions keep the probability of any false alarm over the whole horizon below the nominal level
- ▣ Both address the weaknesses seen above: dependence on the forecast distribution and the one-shot design of the traffic light

Recap: Backtesting ES

- ES tests use only the tail days: expect low power over one year
- McNeil–Frey tests the mean residual, Z_2 tests frequency and size, Du–Escanciano tests the depth of violations
- Z_2 simulation and PIT-based tests need the predictive distribution; residual-bootstrap and regression-based tests avoid specifying it in full
- Multinomial tests at several VaR levels are simple, powerful implicit ES tests

Testing Is Not Ranking

- ▣ A backtest asks: is this model **acceptable**? Several models can pass, or all can fail
- ▣ Comparative backtesting asks: which model is **better**? (Nolde & Ziegel (2017))
- ▣ Ranking needs a **scoring function** $S(x, L)$: the loss of reporting x when L occurs
- ▣ Compare the average scores $\bar{S} = \frac{1}{T} \sum_t S(x_t, L_t)$: lower is better
- ▣ But which scoring function? A bad one can reward a wrong forecast

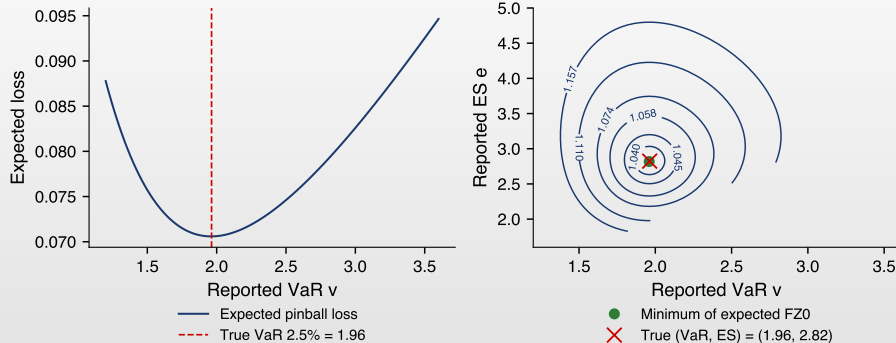
Elicitability

- A statistic $\theta(F)$ is **elicitable** if some scoring function is minimised in expectation exactly by the true value (Gneiting (2011))
- $\theta(F) = \arg \min_x E_F [S(x, L)]$
- VaR: pinball loss $S(v, L) = (1\{L > v\} - \alpha)(L - v)$. Consistency in one line
 - ▶ $\frac{\partial}{\partial v} E_F[S(v, L)] = -P(L > v) + \alpha = F(v) - (1 - \alpha) = 0 \Rightarrow v = F^{-1}(1 - \alpha) = \text{VaR}_\alpha$, and $\frac{\partial^2}{\partial v^2} = f(v) > 0$
- Osband's principle (Fissler & Ziegel (2016)): consistent scores are built from an identification function, here $V(v, L) = 1\{L \leq v\} - (1 - \alpha)$ with $E_F[V] = 0$ only at the truth
- ES is **not** elicitable on its own: no scoring function of e alone is minimised by the true ES (Gneiting (2011))

VaR and ES Together: Fissler and Ziegel (2016)

- The pair (VaR, ES) is jointly elicitable: [Fissler & Ziegel \(2016\)](#) characterise the whole family of consistent scoring functions
- FZ0 loss ([Patton, Ziegel & Chen \(2019\)](#)), with losses positive and $0 < v \leq e$
 - ▶
$$\text{FZ0}(v, e; L) = \frac{1\{L > v\}(L - v)}{\alpha e} + \frac{v}{e} + \ln e - 1$$
- First term: penalty for breaches, scaled by ES; last terms: penalty for reporting a large ES
- Differences of FZ0 losses do not depend on the units of the losses, so they can be compared across assets
- Lower average FZ0 = better joint VaR and ES forecast

Elicitability in Pictures



□ Student-t losses with $\nu = 4$ and unit variance, $\alpha = 2.5\%$; expected losses computed on 2 million draws

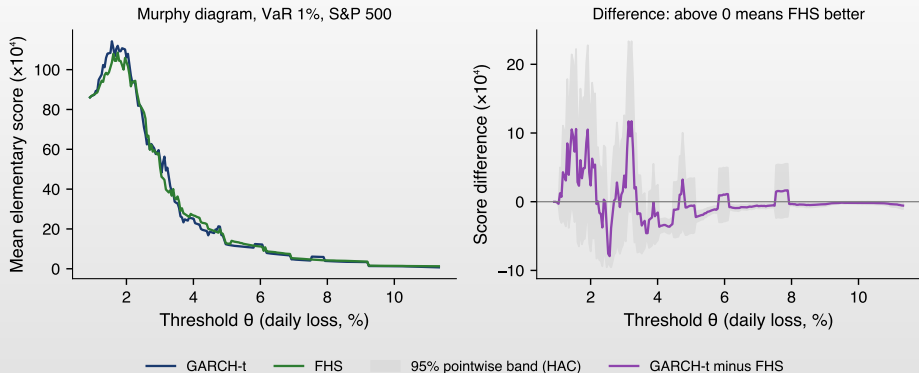
Elicitability: Interpretation

- Left: the expected pinball loss is lowest at the true VaR 2.5%, 1.96
- Right: the expected FZ0 loss is lowest at (1.96, 2.82), i.e. at the true pair (VaR, ES) up to the grid resolution
- A forecaster who knows the truth has no incentive to report anything else: the score is **consistent**
- Consistency is what makes ranking by average score meaningful

Is the Ranking Robust to the Score? Elementary Scores

- For a quantile there are infinitely many consistent scores: $S(v, L) = (1\{L \leq v\} - \tau)(g(v) - g(L))$, any increasing g , $\tau = 1 - \alpha$
- Each is a mixture of **elementary scores** (Ehm, Gneiting, Jordan & Krüger (2016))
 - ▶ $S_\theta(v, L) = (1\{L < v\} - \tau)(1\{\theta < v\} - 1\{\theta < L\})$, $\theta \in \mathbb{R}$; the pinball loss is $\int S_\theta d\theta$
- Forecast A beats B under **every** consistent score iff $\bar{S}_\theta(A) \leq \bar{S}_\theta(B)$ for all θ
- **Murphy diagram**: plot $\bar{S}_\theta$ against θ ; crossing curves mean the ranking depends on the chosen score

Murphy Diagram: GARCH-t versus FHS



- S&P 500, VaR 1%, 2007–2026; θ from 0.9% to 11.3% (range of the forecasts); right: difference with a 95% pointwise HAC band

Murphy Diagram: Interpretation

- ▣ Pinball loss ($\times 10^4$): GARCH-t 3.849, FHS 3.849: a tie under the usual score
- ▣ Below $\theta = 5\%$ the curves cross repeatedly: FHS is better on 50% of thresholds and the band contains zero on 82% of them
- ▣ Above 5% GARCH-t has the lower score on 89% of thresholds: scores that weight crisis thresholds favour it
- ▣ Neither forecast dominates: the ranking is score-specific, so state the score before looking at the results

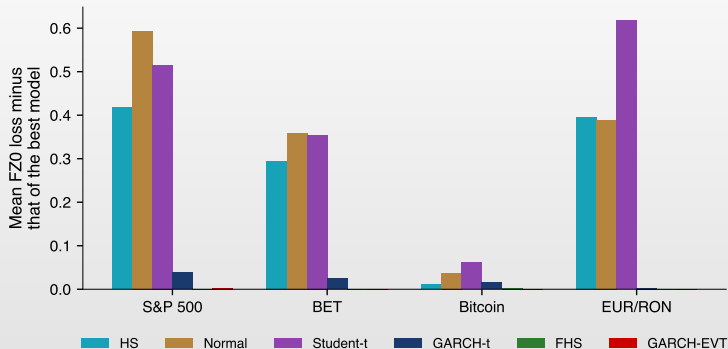
Recap: Elicitability

- Elicitable = there is a scoring function that rewards honesty
- VaR: pinball loss; ES alone: impossible; (VaR, ES) jointly: FZ family, e.g. FZ0
- Elicitability matters for **ranking** models; ES can still be **tested** (Section 4)
- Murphy diagrams check whether a ranking holds for the whole family of consistent scores

The Diebold–Mariano Test on FZ0 Losses

- Loss differential of models A and B : $d_t = \text{FZ0}_{A,t} - \text{FZ0}_{B,t}$
- H_0 : $E[d_t] = 0$ (equal predictive accuracy), Diebold & Mariano (1995)
- $$\text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{LRV}}(d_t)/T}} \xrightarrow{d} N(0, 1)$$
- $\widehat{\text{LRV}}$: long-run variance with Newey–West weights (Newey & West (1987)), 9 lags; tail losses cluster, so d_t is autocorrelated
- $\text{DM} > 0$: model A has the larger average loss (worse); $|\text{DM}| > 1.96$: significant at 5%

Average FZ0 Loss by Model

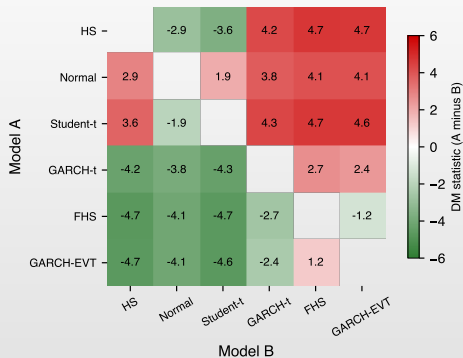


□ Mean FZ0 loss of the (VaR 2.5%, ES 2.5%) forecasts minus that of the best model in each market

FZ0 Ranking: Interpretation

- ▣ Best model: FHS on the S&P 500; GARCH-EVT on BET, Bitcoin and EUR/RON
- ▣ Unconditional models lose by 0.3–0.6 FZ0 units on S&P 500, BET and EUR/RON
- ▣ On Bitcoin all models are close (largest gap 0.062): the data cannot separate them
- ▣ GARCH-t is behind FHS on the S&P 500 by 0.039: small, but is it significant?

Pairwise Diebold–Mariano Tests, S&P 500



□ DM statistic of model A (row) minus model B (column) on FZ0 losses; red: A worse, green: A better

Diebold–Mariano: Interpretation

- HS versus FHS: $DM = 4.70$; GARCH-t versus FHS: $DM = 2.65$: both significantly worse
- GARCH-EVT versus FHS: $DM = 1.24$: statistically indistinguishable
- HS beats the Normal distribution ($DM = 2.88$ for Normal minus HS): a slow empirical quantile is still better than a thin tail
- Many pairwise tests raise a multiple-testing problem: which models survive **jointly**?

Comparing Forecasting Methods: Giacomini–White

- DM targets the population accuracy of **models**; with rolling 1000-day windows and re-estimated parameters we compare **forecasting methods**, and estimation error never vanishes
- Giacomini & White (2006): $H_0: E[\Delta L_t | \mathcal{F}_{t-1}] = 0$ for $\Delta L_t = \text{FZ0}_{A,t} - \text{FZ0}_{B,t}$; valid for nested models and finite windows
- Test function $h_{t-1} = (1, \Delta L_{t-1})'$, $Z_t = h_{t-1} \Delta L_t$: $\text{GW} = T \bar{Z}' \hat{\Omega}^{-1} \bar{Z} \xrightarrow{d} \chi_2^2$, $\hat{\Omega} = T^{-1} \sum_t Z_t Z_t'$ (one step ahead)
- With $h_{t-1} = 1$ the statistic has the DM form: the DM numbers above stay valid, now read as tests on methods
- S&P 500 against FHS: GARCH-EVT $\text{GW} = 7.38$ ($p = 0.025$), GARCH-t 7.59 ($p = 0.022$), HS 53.38 ($p < 0.001$); GARCH-EVT, a tie under DM, is rejected conditionally: the lagged differential predicts which method wins tomorrow

Three-Zone Comparative Backtesting (Nolde and Ziegel)

- Internal model A against a standard model S (here HS), mean FZ0 loss; two one-sided tests at 5% (Nolde & Ziegel (2017))
- H_0^- : A at least as good as S ; H_0^+ : A at most as good. **Green**: reject H_0^+ ; **red**: reject H_0^- ; **yellow**: neither
- The burden of proof is reversed: the bank must show that its model is better, not the supervisor that it is worse

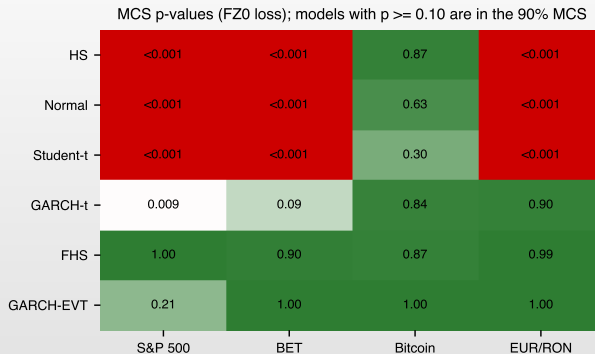
DM (A minus HS)	S&P 500	BET	Bitcoin	EUR/RON
GARCH-t	-4.23	-5.18	0.09	-6.66
FHS	-4.70	-5.49	-0.17	-6.94
GARCH-EVT	-4.68	-5.53	-0.25	-6.92

- Colour = zone. Green everywhere except Bitcoin, where every model is yellow: with its noisy tail, no model can prove it beats HS

The Model Confidence Set

- ▣ Hansen, Lunde & Nason (2011): a set of models built to contain the best models with asymptotic probability at least $1 - \alpha_{MCS}$, under the method's regularity conditions
- ▣ Algorithm
 - ▶ Test H_0 : all models in the current set have equal expected loss
 - ▶ If rejected, eliminate the model with the largest standardised average loss and repeat
 - ▶ Critical values from a block bootstrap (blocks of 10 days, 1000 replications)
- ▣ MCS p -value of a model: the largest test p -value up to its elimination; keep models with $p \geq 0.10$
- ▣ A large MCS is an honest answer: the data do not separate the models

Which Models Survive?



□ MCS p -values for the FZ0 loss of the (VaR 2.5%, ES 2.5%) forecasts, per market

MCS: Interpretation

- ▣ S&P 500: only FHS and GARCH-EVT (MCS p 0.21) remain; GARCH-t is excluded (p 0.01)
- ▣ EUR/RON: the three conditional models remain; BET: FHS and GARCH-EVT remain, GARCH-t just misses (p 0.09)
- ▣ Unconditional models are excluded everywhere except Bitcoin
- ▣ Bitcoin: all six models remain: the data cannot separate them (compare [Pele & Mazurencu-Marinescu-Pele \(2026\)](#))
- ▣ Conclusion across tests: FHS and GARCH-EVT are the models to beat

Recap: Comparative Backtesting

- ▣ Rank models with a consistent score (FZ0 for VaR and ES); compare them with DM tests using HAC variances
- ▣ With estimated rolling-window forecasts, read the tests as Giacomini–White tests on methods; conditioning can overturn an unconditional tie
- ▣ Regulatory use: the three-zone comparative test puts the burden of proof on the bank
- ▣ The MCS handles many models at once and shows which ones the data cannot separate (non-rejection is not proof of equal performance)
- ▣ A model can pass all backtests and still be dominated by a better one

A Distribution-Free Promise



Emmanuel Candès, co-author of adaptive conformal inference

Photo: Cmicel67 (2026); CC BY-SA 4.0;
Wikimedia Commons

- All models so far assume a distribution (Normal, t , GPD) or a stable history, as HS does
- **Conformal prediction** turns any forecast into a quantile with a coverage guarantee (Lei et al. (2018), Angelopoulos & Bates (2021))
- The only assumption: the data are **exchangeable**, i.e. their joint distribution does not change under reordering
- Financial returns have volatility clustering: exchangeability is exactly what fails

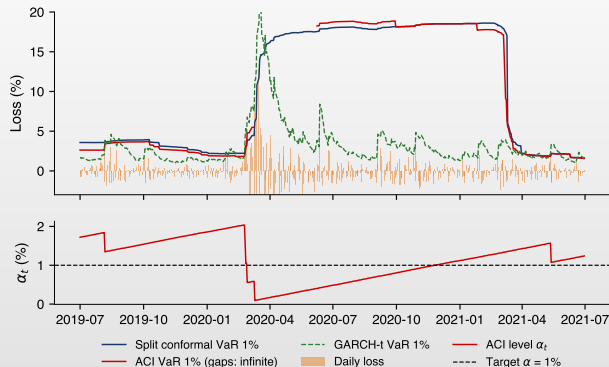
Split Conformal Quantiles

- Nonconformity score: $s_t = L_t / \hat{\sigma}_t$, with $\hat{\sigma}_t$ the standard deviation of the previous 250 returns
- Calibration set: the last $n = 250$ scores; sort them $s_{(1)} \leq \dots \leq s_{(n)}$
- Conformal VaR: $\text{VaR}_t = \hat{\sigma}_t s_{(k)}$ with $k = \lceil (n+1)(1-\alpha) \rceil$
- Guarantee if the $n+1$ scores are exchangeable: $P(L_t \leq \text{VaR}_t) \geq 1 - \alpha$ (and $\leq 1 - \alpha + \frac{1}{n+1}$ without ties)
 - ▶ Without exchangeability the coverage gap is bounded by weighted total-variation distances between the joint vector of scores and the same vector with the test score swapped with calibration score i (Barber, Candès, Ramdas & Tibshirani (2023))
- Example: $n = 250$, $\alpha = 0.01$: $k = \lceil 251 \cdot 0.99 \rceil = 249$, the second largest score; breach probability between 0.60% and 1%

Adaptive Conformal Inference

- Gibbs & Candès (2021): update the working level after each day, $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, $\text{err}_t = 1\{L_t > \text{VaR}_t\}$
- After a breach α_t falls (VaR rises); after calm days it slowly rises again; we use $\gamma = 0.005$
 - Choosing γ : Zaffran et al. (2022) aggregate ACI experts over a grid of γ values
- Guarantee without any assumption on the data: $\left| \frac{1}{T} \sum_t \text{err}_t - \alpha \right| \leq \frac{\max(\alpha_1, 1 - \alpha_1) + \gamma}{\gamma T}$
- The guarantee is on the long-run **average**, not on each day or each crisis
- If $\alpha_t < 1/(n+1)$, the required quantile does not exist: the VaR becomes infinite

Conformal VaR Through COVID-19

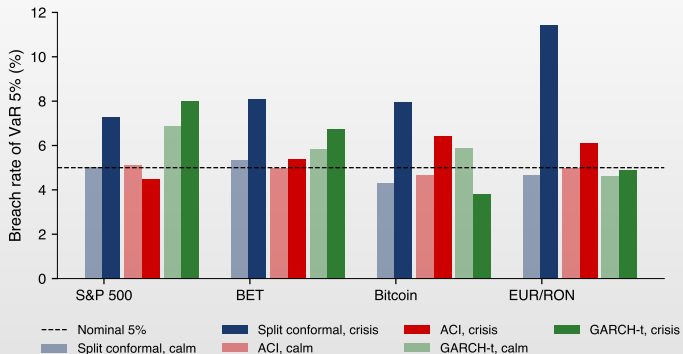


□ S&P 500, July 2019–June 2021: split conformal, ACI and GARCH-t VaR 1% (top) and the ACI level α_t (bottom)

Conformal VaR in a Crisis: Interpretation

- 20.02–30.04.2020 (50 days, 0.5 breaches expected): split conformal 6 breaches, ACI 4
- ACI set $\text{VaR} = \infty$ on 37 of these 50 days: this stopped further breaches but missed the target (4 breaches, 8% of days); an infinite VaR carries no information
- After the crash, the calibration set contains March 2020: the split conformal VaR stays near 18% for a whole year
- GARCH-t adapts in days because it models the dependence; split conformal updates its scale and scores daily, but its 250-day windows react slowly; the ACI level moves only after breaches

Coverage in Calm and Crisis Periods



- Breach rates of the VaR 5%; crisis days: volatility of the previous 20 days above the 90th percentile of its values over the previous 1000 days (known in advance)

Calm Versus Crisis: Interpretation

- Over all days split conformal is close to 5%: S&P 500 5.34%, BET 5.63%, Bitcoin 4.58%, EUR/RON 5.06%
- After turbulent periods it breaches more: S&P 500 7.29%, BET 8.08%, Bitcoin 7.95%, EUR/RON 11.43%
- ACI brings the crisis rates close to 5% on equities (S&P 500 4.49%, BET 5.38%) and lowers them on Bitcoin (6.44%) and EUR/RON (6.12%)
- For VaR 1% ACI keeps the average only by an infinite VaR on 13.4% of S&P 500 days
- Overall rates are close to nominal, crisis rates are not; a promising route is to conformalise a good conditional model (scores standardised by GARCH volatility)

Recap: Conformal Prediction

- Split conformal: finite-sample coverage under exchangeability, with no distributional model
- ACl: long-run average coverage for any sequence, possibly with infinite VaR
- Both fail conditionally in volatility clusters: backtest them like any other model

Research Question: Can a Supervisor Backtest ES on Its Own?

- Bayer & Dimitriadis (2022) (accepted version: [arXiv:1801.04112v2](https://arxiv.org/abs/1801.04112v2)): can ES forecasts be backtested using only the realised returns and the ES forecasts themselves?
- Why it matters: FRTB moves the capital charge to ES 2.5% (Section 3 of this lecture)
 - ▶ The ES tests of Section 4 also need VaR forecasts, the volatility or the whole forecast distribution (Z_2 , McNeil & Frey (2000), Du & Escanciano (2017), Kratz, Lok & McNeil (2018)); supervisors do not receive these inputs (Section 1 of the paper)
 - ▶ A rejection of such a joint test does not say which input is wrong (Definition 2.1: a backtest of a risk measure uses only the returns and the forecasts of that measure)
- Idea: a Mincer–Zarnowitz regression of the returns on the ES forecasts, with H_0 : intercept 0, slope 1
 - ▶ ES alone is not elicitable (Section 5), so no loss function exists for an ES-only regression; the pair (VaR, ES) is jointly elicitable (Fissler & Ziegel (2016)), so a joint quantile–ES regression can be estimated (Patton, Ziegel & Chen (2019))

Data and Method, as Published

- Returns convention: r_t daily log return, $\hat{e}_t = -\widehat{\text{ES}}_t < 0$ the ES forecast in return units, $\tau = 2.5\%$; ES regression (Eq. 2.4–2.6): $r_t = \gamma_1 + \gamma_2 \hat{e}_t + u_t^e$, with $\text{ES}_\tau(u_t^e | \mathcal{F}_{t-1}) = 0$
 - ▶ **Auxiliary ESR** (Eq. 2.8–2.10): quantile equation on $(1, \hat{v}_t)$, Wald test of $(\gamma_1, \gamma_2) = (0, 1)$; needs VaR forecasts too
 - ▶ **Strict ESR** (Eq. 2.11–2.13): quantile equation also on $(1, \hat{e}_t)$: only ES forecasts enter; exact when VaR and ES are proportional (pure scale models)
 - ▶ **Intercept ESR** (Eq. 2.14–2.15): $r_t - \hat{e}_t = \gamma_1 + u_t^e$; one-sided t -test of $H_0: \gamma_1 \geq 0$ against $\gamma_1 < 0$ (ES underestimated)
- Estimation (Eq. 2.16–2.18): M-estimator that minimises the FZ0 loss of Section 5; asymptotic theory under misspecification (Theorem 2.4) and a misspecification-robust sandwich covariance (Section 2.5); R packages [esback](#) and [esreg](#)
- Simulations (Section 3): 10 000 replications, T from 250 to 5 000 days; EGARCH-t calibrated to the S&P 500, AR(1)-GARCH(1,1), two Student-t GAS models; power against HS on 250 days
 - ▶ Competitors: the exceedance-residual test of [McNeil & Frey \(2000\)](#) and the conditional calibration tests of [Nolde & Ziegel \(2017\)](#)
- Application (Section 4): daily log returns of the 200 largest S&P 500 stocks; out-of-sample January 2010 – August 2019 (2 432 days)
 - ▶ GARCH(1,1) and GJR-GARCH(1,1), Normal or Student-t innovations; rolling windows of 250, 500, 1 000, 1 500, 2 000 days; refit every 5, 21, 62, 125, 250 days; one-sided Intercept ESR at 5%

Key Results

Finding	Source	Reading
Size at nominal 5%, EGARCH-t: Strict ESR 0.09 ($T = 250$) and 0.05 ($T = 1000$) with the robust covariance; 0.24 and 0.11 with the classical one	Table 1	The robust covariance is what makes the test usable in short samples
Strict and Auxiliary ESR sizes differ by at most 0.01 for every process and sample size, including AR(1)-GARCH with $\phi = 0.5$	Table 1	Dropping the VaR forecasts costs almost nothing: ES can be tested on its own
Simple conditional calibration test: size 0.29 at $T = 250$ and 0.14 at $T = 1000$ (EGARCH-t)	Table 1	A competitor with high raw power can owe it to over-rejection
Exceedance-residual tests barely react to wrong volatility dynamics or to forecasts for a wrong level $\tilde{\tau} \neq \tau$; ESR tests detect every misspecification	Fig. 3	A bank could report ES for a milder level and pass the older tests
Normal innovations: rejected for 94%–100% of the stocks in every window and refit design	Table 3	Normal tails understate ES for almost every stock
Student-t: GARCH 0.59 with a 250-day window and yearly refit; 0.08–0.10 with 1500 or 2000 days and refit at least quarterly, for GARCH and GJR-GARCH	Table 3	Fat tails, more than four years of data and quarterly refits are enough; the GJR-GARCH leverage term adds nothing

Replicate It, Limits and an Open Extension

- Replication with the course data
 - ▶ Run the Strict ESR (two-sided) and the Intercept ESR (one-sided) on the ES 2.5% forecasts of this lecture's six models: S&P 500, BET, Bitcoin (data/market) and EUR/RON (BNR); compare the decisions with Z_2 and McNeil–Frey (Section 4)
 - ▶ Repeat the Table 3 grid for GARCH(1,1) with Normal and Student-t innovations on the S&P 500: does the Normal model fail and the Student-t model pass?
- Limits stated by the authors
 - ▶ Strict and Intercept ESR: the quantile equation is misspecified when VaR and ES are not proportional (Section 2.2); Auxiliary and Strict ESR are two-sided only (Section 2.3)
 - ▶ The robust covariance assumes a location-scale model for the conditional distribution (Section 2.5); the classical covariance is oversized in small samples (Section 3.1)
- Open extension: a one-year supervisory window
 - ▶ At $T = 250$ the two-sided Intercept ESR has size 0.14 at nominal 5% (Table 1, EGARCH-t); the one-sided size is reported only for $T = 2500$ (0.03, Table 2, GARCH(1,1)-t)
 - ▶ Open: what is the one-sided size at the one-year horizon, and can a bootstrap of the t -statistic hold it at 5% without losing the power of Fig. 4?
- Project seed: Seminar 8, Part C3

An Open Question: Conditional Coverage in Crises

- Distribution-free VaR guarantees are **marginal**: split conformal needs exchangeability, ACI only controls the long-run average
 - ▶ Barber, Candès, Ramdas & Tibshirani (2023): without exchangeability the coverage loss is bounded by total-variation distances between the score vector and its versions with the test score swapped in; how large they are under volatility clustering is an empirical question
 - ▶ Gibbs, Cherian & Candès (2025): exact coverage conditional on a pre-specified class of shifts is attainable; which class captures a financial crisis is open
- Formal question, with a crisis flag $C_t \in \mathcal{F}_{t-1}$ known ex ante
 - ▶ H_0 : $P(L_t > \text{VaR}_t \mid C_t = 1) = \alpha$ against H_1 : $P(L_t > \text{VaR}_t \mid C_t = 1) \neq \alpha$
 - ▶ Falsified by a crisis breach rate outside the Binomial acceptance region, after correcting for dependence between breaches and for multiplicity
- Stakes: capital is needed exactly on the days when marginal guarantees say least

The Discovery Loop, Step by Step (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: map the guarantees, then check every DOI on [Crossref](#) (the title must match)
 - ▶ List peer-reviewed work since 2019 on conformal prediction under serial dependence; for each, give the DOI, the exact guarantee (marginal, conditional or long-run) and its assumptions.
- 2. **Competing hypotheses**, each with a testable implication
 - ▶ Give three rival explanations for why split conformal VaR over-breaches after volatile periods (stale scale, changing tail shape, regime-selection artefact) and what each predicts for breach rates by regime.
- 3. **Data and code**: replicate a known number before changing anything
 - ▶ Write Python that reads data/market/GSPC.INDX.csv, builds the split conformal VaR 5% of this chapter with strictly lagged inputs and prints the crisis breach rate. Target: 7.29%

The Discovery Loop, Step by Step (2/2)

- 4. **Robustness**, written down before the results are seen
 - ▶ Draft a pre-registered robustness plan: `alpha in {1%, 5%}`, `calibration size in {125, 250, 500}`, `crisis cut-off at the 80th, 90th and 95th percentile`, `two subsamples`, `Holm correction across markets`.
 - ▶ Placebo: random crisis labels with the same frequency give crisis breach rates centred on the full-sample rate; their permutation distribution is the reference band
- 5. **Critique**: ask the assistant to attack the result
 - ▶ Act as a hostile referee: `list the strongest reasons this crisis effect is an artefact (clustered breaches, regime defined from the same data, overlapping windows)` and the test that would settle each.
- 6. **Report**: every specification tried, effect sizes with intervals, code, and the log of AI use

What the Researcher Must Check

- **Look-ahead:** the scale $\hat{\sigma}_t$, the GARCH parameters and the crisis flag may use only $\mathcal{F}_{t-1}$; test: recompute with every input lagged one more day
- **Dependent breaches:** clustered hits make Binomial p -values too small; test: Christoffersen and duration tests, or a block bootstrap of the hit sequence
- **Power:** S&P 500 has 613 crisis days, i.e. 6.1 expected breaches of VaR 1%; a true rate of 2% is detected with probability 57%; compute power before testing
- **Multiple testing:** 4 markets $\times$ 2 levels $\times$ 3 methods = 24 tests; control the FWER (Holm) or the FDR (BH)
- **Misstated theorems and invented references:** e.g. “ACI guarantees conditional coverage”; check the theorem in [Gibbs & Candès \(2021\)](#) and every DOI
- **Off-by-one quantiles:** the index $k = \lceil (n+1)(1-\alpha) \rceil$ is not the default of `np.quantile`; test on simulated i.i.d. data

Mini-Case: Does a GARCH Scale Repair Crisis Coverage?

- Replication: the split conformal VaR 5% of this chapter breaches on 7.29% of S&P 500 crisis days (Section 7)
- A typical assistant proposal: score $s_t = (L_t + \mu_t)/\sigma_t$ with the GARCH-t forecasts; H_0 : crisis breach rate = 5%, exact Binomial test, Holm across markets

Market	Crisis days	Split conformal	Holm p	GARCH-scaled	Holm p
S&P 500	613	7.01%	0.052	2.77%	0.037
BET	506	8.10%	0.009	3.95%	0.309
Bitcoin	150	8.00%	0.092	1.33%	0.111
EUR/RON	245	11.43%	<0.001	2.45%	0.153

- Result: the GARCH scale pushes every crisis rate below 5%; Holm rejects only on the S&P 500 (2.77%); elsewhere this is non-rejection, not proof of coverage: Binomial p -values ignore clustered breaches and crisis samples are small; next hypothesis: the GARCH scale overshoots after spikes
- Common sample: 4 709 S&P 500 days after 250 calibration days; code: `Quantlets/Ch_08/ai_discovery_case.py`

Your Turn: A Project Seed

- Question: can a distribution-free VaR 1% and ES 2.5% forecast achieve crisis-conditional coverage in all four markets under a pre-registered test?
- Replicate first: the crisis breach rates of split conformal VaR 5% (S&P 500 7.29%, EUR/RON 11.43%) and the ACI share of infinite VaR 1% (13.4%)
- Deliverables
 - ▶ A pre-registration (hypotheses, tests, corrections, power) dated before the first run
 - ▶ A candidate method, e.g. [Gibbs, Cherian & Candès \(2025\)](#) with the volatility regime as the conditioning class; VaR: Kupiec, Christoffersen and duration tests by regime
 - ▶ ES: quantile coverage does not pin down the tail mean; state how ES is produced and test it with the ES regression backtest ([Bayer & Dimitriadis \(2022\)](#)), or with Z_2 if the method yields a forecast distribution
 - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in `AI_USE.md` and every error the assistant made in `AI_ERRORS.md`: github.com/danpele/MFM/tree/main/project

Scorecard Across Four Markets

- Each cell: breach rate of the VaR 1% in % / Z_2 p -value / MCS p -value (FZ0)

Model	S&P 500	BET	Bitcoin	EUR/RON
HS	1.67 / <0.001 / <0.001	1.17 / 0.134 / <0.001	0.77 / 0.863 / 0.87	0.51 / 1.000 / <0.001
Normal	2.70 / <0.001 / <0.001	1.90 / <0.001 / <0.001	1.74 / 0.001 / 0.63	1.06 / 0.973 / <0.001
Student-t	2.00 / 0.002 / <0.001	1.58 / 0.003 / <0.001	0.71 / 0.625 / 0.30	0.85 / 0.920 / <0.001
GARCH-t	1.77 / <0.001 / 0.01	1.46 / <0.001 / 0.09	1.12 / 0.050 / 0.84	1.17 / 0.630 / 0.90
FHS	1.23 / 0.172 / 1.00	1.20 / 0.188 / 0.90	0.92 / 0.345 / 0.87	0.85 / 0.968 / 0.99
GARCH-EVT	1.13 / 0.090 / 0.21	0.99 / 0.210 / 1.00	0.95 / 0.581 / 1.00	0.78 / 0.976 / 1.00

- Only FHS and GARCH-EVT pass the Z_2 test at 5% and stay in the 90% MCS in all four markets

A Romanian Bank's Trading Book



National Bank of Romania, Bucharest

Photo: Britchi Mirela (2018); CC BY-SA 4.0; Wikimedia

[Commons](#)

- Typical exposures: Romanian equities, via the BET index, and the EUR/RON exchange rate
- BET: HS and the Normal distribution fail the duration tests; GARCH-EVT passes all VaR and ES tests
- EUR/RON: long calm spells and rare jumps; HS breaches only 0.51% of days, and all unconditional models fail the independence test
- Seminar 8, Part C: which model would you certify for this book?

From Our Research Group

- ▣ Pele, Lazar & Dufour (2017): entropy of intraday returns predicts the daily VaR of EUR/JPY
- ▣ Pele & Mazurencu-Marinescu-Pele (2019): entropy-based VaR forecasts for Bitcoin, backtested with Christoffersen's tests
- ▣ Pele, Lazar & Mazurencu-Marinescu-Pele (2019): a tail-entropy measure of ES and its backtesting performance
- ▣ Pele & Mazurencu-Marinescu-Pele (2026): when are two ES forecasts statistically distinguishable?
- ▣ Textbooks: Franke, Härdle & Hafner (2019) (VaR and backtesting chapters) and Härdle, Chen & Overbeck (2017)

Further Study on Quantinar

Advanced course on quantinar.com

 Machine learning in Financial Risk

Code and data

-  All charts and tests: github.com/danpele/MFM/tree/main/Quantlets/Ch_08
-  Series: S&P 500, BET, Bitcoin (repository folder `data/market`); EUR/RON: BNR reference rate

Key Takeaways

- ▣ Test the number and the timing of breaches (coverage, independence, durations, DQ), and know the size and power of the tests at your T
- ▣ Regulators use simple counts (traffic light), now combined with ES-based capital under FRTB
- ▣ ES can be backtested (Z_2 , residuals, cumulative violations) but not ranked alone: use FZ0 for (VaR, ES)
- ▣ Rank with DM/Giacomini–White tests and the MCS, check robustness with Murphy diagrams; accept that some models cannot be separated
- ▣ Conformal guarantees are marginal: they do not protect you in the crisis
- ▣ Empirically, a GARCH filter plus a flexible tail (FHS, EVT) wins in all four markets

References (1/5)

- ▣ Acerbi, C., Szekely, B. (2014). [Backtesting expected shortfall](#). MSCI Inc., December 2014.
- ▣ Angelopoulos, A.N., Bates, S. (2021). [A gentle introduction to conformal prediction and distribution-free uncertainty quantification](#). [arXiv:2107.07511](#).
- ▣ Barber, R.F., Candès, E.J., Ramdas, A., Tibshirani, R.J. (2023). [Conformal prediction beyond exchangeability](#). *Annals of Statistics*, 51(2).
- ▣ Basel Committee on Banking Supervision (1996). [Supervisory framework for the use of “backtesting” in conjunction with the internal models approach to market risk capital requirements](#). Bank for International Settlements.
- ▣ Basel Committee on Banking Supervision (2019). [Minimum capital requirements for market risk](#) (rev. February 2019). Bank for International Settlements.
- ▣ Bayer, S., Dimitriadis, T. (2022). [Regression-based expected shortfall backtesting](#). *Journal of Financial Econometrics*, 20(3), 437–471. Accepted version: [arXiv:1801.04112v2](#).
- ▣ Christoffersen, P.F. (1998). [Evaluating interval forecasts](#). *International Economic Review*, 39(4), 841–862.
- ▣ Christoffersen, P., Pelletier, D. (2004). [Backtesting value-at-risk: a duration-based approach](#). *Journal of Financial Econometrics*, 2(1), 84–108.

References (2/5)

- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Du, Z., Escanciano, J.C. (2017). Backtesting expected shortfall: accounting for tail risk. *Management Science*, 63(4), 940–958.
- ▣ Ehm, W., Gneiting, T., Jordan, A., Krüger, F. (2016). Of quantiles and expectiles: consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society, Series B*, 78(3), 505–562.
- ▣ Engle, R.F., Manganelli, S. (2004). CAViaR: conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Escanciano, J.C., Olmo, J. (2010). Backtesting parametric value-at-risk with estimation risk. *Journal of Business & Economic Statistics*, 28(1), 36–51.
- ▣ Fissler, T., Ziegel, J.F. (2016). Higher order elicibility and Osband's principle. *Annals of Statistics*, 44(4).
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.

References (3/5)

- ▣ Gibbs, I., Candès, E. (2021). [Adaptive conformal inference under distribution shift](#). *Advances in Neural Information Processing Systems*, 34; arXiv:2106.00170.
- ▣ Gibbs, I., Cherian, J.J., Candès, E.J. (2025). [Conformal prediction with conditional guarantees](#). *Journal of the Royal Statistical Society, Series B*, 87(4), 1100–1126.
- ▣ Gneiting, T. (2011). [Making and evaluating point forecasts](#). *Journal of the American Statistical Association*, 106(494), 746–762.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). [The model confidence set](#). *Econometrica*, 79(2), 453–497.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (eds.) (2017). [Applied Quantitative Finance](#), 3rd ed. Springer.
- ▣ Hoga, Y., Demetrescu, M. (2023). [Monitoring value-at-risk and expected shortfall forecasts](#). *Management Science*, 69(5), 2954–2971.
- ▣ Kratz, M., Lok, Y.H., McNeil, A.J. (2018). [Multinomial VaR backtests: a simple implicit approach to backtesting expected shortfall](#). *Journal of Banking & Finance*, 88, 393–407.
- ▣ Kupiec, P.H. (1995). [Techniques for verifying the accuracy of risk measurement models](#). *Journal of Derivatives*, 3(2), 73–84.

References (4/5)

- ▣ Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R.J., Wasserman, L. (2018). Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523), 1094–1111.
- ▣ McNeil, A.J., Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Nolde, N., Ziegel, J.F. (2017). Elicitability and backtesting: perspectives for banking regulation. *Annals of Applied Statistics*, 11(4).
- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). Dynamic semiparametric models for expected shortfall (and value-at-risk). *Journal of Econometrics*, 211(2), 388–413.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). Information entropy and measures of market risk. *Entropy*, 19(5), 226.
- ▣ Pele, D.T., Lazar, E., Mazurencu-Marinescu-Pele, M. (2019). Modeling expected shortfall using tail entropy. *Entropy*, 21(12), 1204.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily value at risk. *Entropy*, 21(2), 102.

References (5/5)

- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). [Finite-sample precision limits for expected shortfall forecast comparisons](#). *Mathematics*, 14(13), 2316.
- ▣ Zaffran, M., Féron, O., Goude, Y., Josse, J., Dieuleveut, A. (2022). [Adaptive conformal predictions for time series](#). *Proceedings of the 39th International Conference on Machine Learning*; arXiv:2202.07282.